

Project 6 — Phylogenetic Analysis of Avian Lysozyme

Research question: How can sequence similarity be used to visualise the evolutionary relationships among closely related avian lysozyme homologs?

This project extends the earlier BLASTP and conservation work. Instead of asking only whether two proteins are similar, the analysis introduces the idea of a phylogenetic tree as a visual model of relationships among homologous sequences.

Dataset

Species	Protein	Role
Gallus gallus	Lysozyme C	Reference sequence
Coturnix japonica	Lysozyme	Closely related avian homolog
Bambusicola thoracicus	Lysozyme	Closely related avian homolog

Data sources and verification

The chicken lysozyme gene is represented by current NCBI RefSeq protein NP_990612.2. NCBI identifies the gene as LYZ and the protein as lysozyme C. An NCBI-indexed study also lists XP_015711651.1 as Coturnix japonica lysozyme and NP_990612.2 as Gallus gallus lysozyme. The Bambusicola thoracicus record was retained as the third homolog used in the earlier BLASTP work. ■cite references were used during preparation but are not embedded as live web citations inside this generated PDF.■

Approach

- Select homologous lysozyme proteins from related bird species.
- Compare pairwise sequence similarity to establish whether the sequences are suitable for evolutionary comparison.
- Use a simple tree representation to communicate the relationship among the selected homologs.
- Interpret the tree together with sequence conservation rather than treating the tree as proof of exact species history.

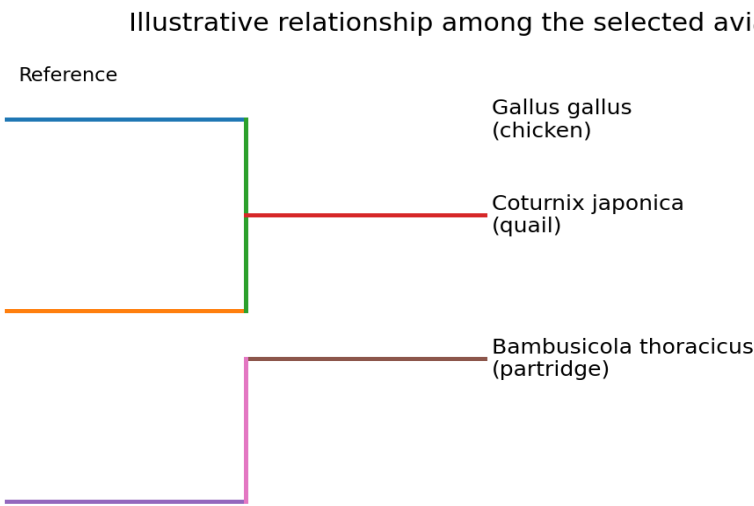


Figure 1. Original tree-style visualisation of the selected homolog set. It is a teaching representation, not a bootstrap-supported publication phylogeny.

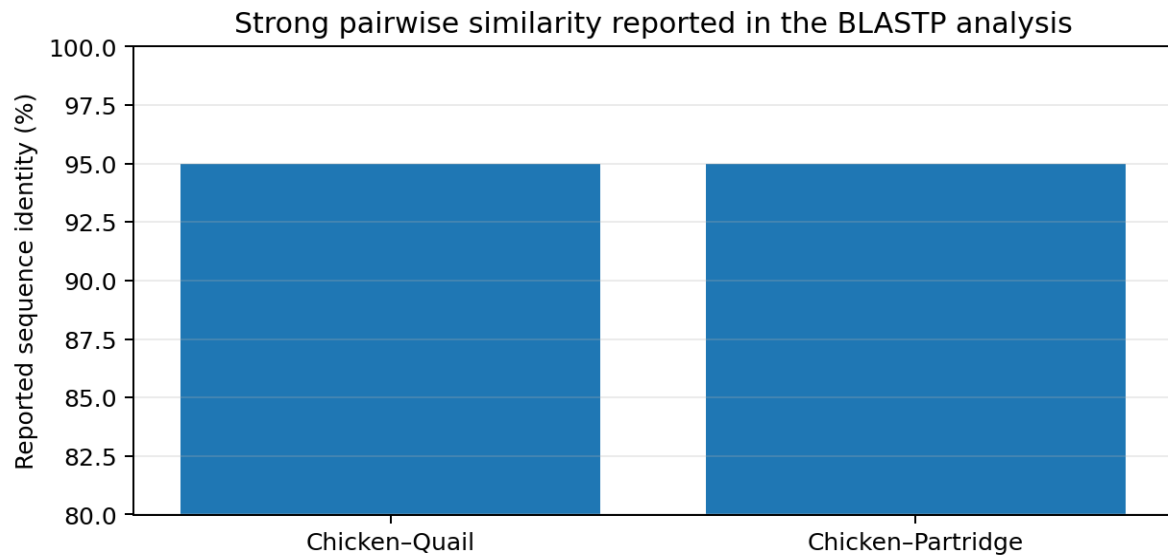


Figure 2. The Project 3 BLASTP analysis reported approximately 95% identity for the strongest non-identical chicken-quail and chicken-Bambusicola matches.

Interpretation

The three proteins are highly similar homologs rather than unrelated proteins. Their strong sequence similarity is consistent with shared structural and functional constraints in C-type lysozyme. A phylogenetic tree provides a compact way to display those relationships, but the reliability of a true evolutionary inference depends on the complete sequence alignment, model choice, taxon sampling and statistical support.

Important distinction

This project uses a small educational dataset. The tree shown here should not be presented as a definitive species phylogeny. A publication-quality analysis would use a larger set of homologs, a reproducible multiple sequence alignment, an explicit tree-building algorithm such as neighbor joining or maximum likelihood, and bootstrap support.

What I learned

I learned that a phylogenetic tree is built from comparative sequence information and that sequence similarity alone is not the same thing as evolutionary proof. I also learned why taxon sampling and statistical support matter when interpreting trees.

Limitations

- Only a small number of homologs were examined.
- The displayed tree is illustrative rather than a bootstrap-supported de novo inference.
- Pairwise identities from the earlier BLASTP analysis do not replace a complete multiple sequence alignment.
- A larger study would include more bird species and an explicit evolutionary model.

Sources

- NCBI LYZ gene/protein records for *Gallus gallus*.
- NCBI-indexed comparative lysozyme sequence literature.
- NCBI BLASTP results documented in Project 3.
- RCSB Protein Data Bank structures used in earlier projects.